

17

Elimination reactions

Connections

➡ Building on

- Stereochemistry [ch14](#)
- Mechanisms of nucleophilic substitution at saturated carbon [ch15](#)
- Conformation [ch16](#)

Arriving at

- Elimination reactions
- What factors favour elimination over substitution
- The three important mechanisms of elimination reactions
- The importance of conformation in elimination reactions
- How to use eliminations to make alkenes (and alkynes)

➡ Looking forward to

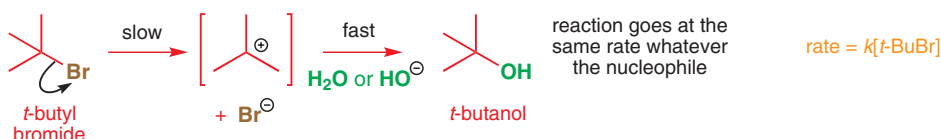
- Electrophilic additions to alkenes (the reverse of the reactions in this chapter) [ch19](#)
- How to control double-bond geometry [ch27](#)

➡ Remember the turnstiles at the railway station (see p. 332).

Substitution and elimination

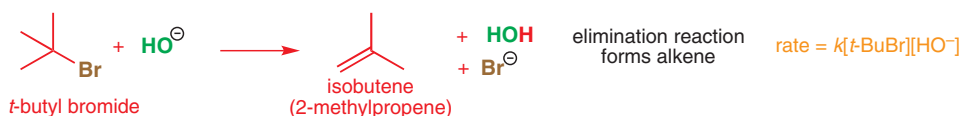
Substitution reactions of *t*-butyl halides, you will recall from Chapter 15, invariably follow the S_N1 mechanism. In other words, the rate-determining step of their substitution reactions is unimolecular—it involves only the alkyl halide. This means that, no matter what the nucleophile is, the reaction goes at the same rate. You can't speed this S_N1 reaction up, for example, by using hydroxide instead of water, or even by increasing the concentration of hydroxide. You'd be wasting your time, we said (see p. 332).

nucleophilic substitution reactions of *t*-BuBr



You'd also be wasting your alkyl halide. This is what actually happens if you try the substitution reaction with a *concentrated* solution of sodium hydroxide.

reaction of *t*-BuBr with concentrated solution of NaOH



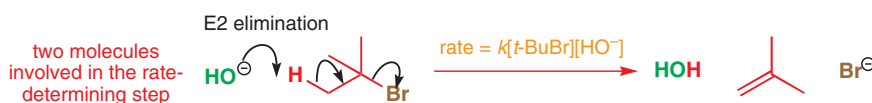
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

The reaction stops being a substitution and an alkene is formed instead. Overall, HBr has been lost from the alkyl halide, and the reaction is called an **elimination**.

In this chapter we will talk about the mechanisms of elimination reactions—as in the case of substitutions, there is more than one mechanism for eliminations. We will compare eliminations with substitutions—either reaction can happen from almost identical starting materials, and you will learn how to predict which is the more likely. Much of the mechanistic discussion relates very closely to Chapter 15, and we suggest that you make sure you understand all of the points in that chapter before tackling this one. This chapter will also tell you about uses for elimination reactions. Apart from a brief look at the Wittig reaction in Chapter 11, this is the first time you have met a way of making simple alkenes.

Elimination happens when the nucleophile attacks hydrogen instead of carbon

The elimination reaction of *t*-butyl bromide happens because the nucleophile is *basic*. You will recall from Chapter 10 that there is *some* correlation between basicity and nucleophilicity: strong bases are usually good nucleophiles. Being a good nucleophile doesn't get hydroxide anywhere in the substitution reaction because it doesn't appear in the first-order rate equation. But being a good base does get it somewhere in the elimination reaction because hydroxide is involved in the rate-determining step of the elimination, and so it appears in the rate equation. This is the mechanism.

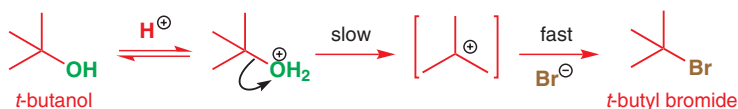


The hydroxide is behaving as a base because it is attacking the hydrogen atom, instead of the carbon atom it would attack in a substitution reaction. The hydrogen atom is not acidic, but proton removal can occur because bromide is a good leaving group. As the hydroxide attacks, the bromide is forced to leave, taking with it the negative charge. Two molecules—*t*-butyl bromide and hydroxide—are involved in the rate-determining step of the reaction. This means that the concentrations of both appear in the rate equation, which is therefore second-order and this mechanism for elimination is termed E2, for *elimination, bimolecular*.

$$\text{rate} = k_2 [t\text{-BuBr}][\text{HO}^-]$$

Now let's look at another sort of elimination. We can approach it again by thinking about another S_N1 substitution reaction, the reverse of the one at the beginning of the chapter: an alcohol is converted into an alkyl halide.

nucleophilic substitution of *t*-BuOH with HBr



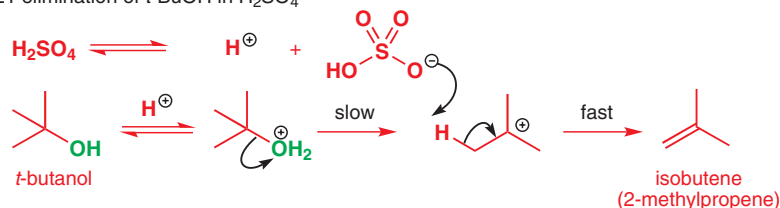
Bromide, the nucleophile, is not involved in the rate-determining step, so we know that the rate of the reaction will be independent of the concentration of Br[−]. Indeed the first step, to form the cation, will happen just as fast even if there is *no bromide at all*. But what happens to the carbocation in such a case? To find out, we need to use an acid whose counterion is such a weak nucleophile that it won't even attack the positive carbon of the carbocation. Here is an example—*t*-butanol in sulfuric acid doesn't undergo substitution, but undergoes elimination instead.

■ The correlation between basicity and nucleophilicity is best for attack at C=O. In Chapter 15 you met examples of nucleophiles that are good at substitution at saturated carbon (such as I[−], Br[−], PhS[−]) but that are not strong bases.

■ Note: No subscripts or superscripts, just plain old E2.

Interactive E1 elimination mechanism

E1 elimination of *t*-BuOH in H₂SO₄



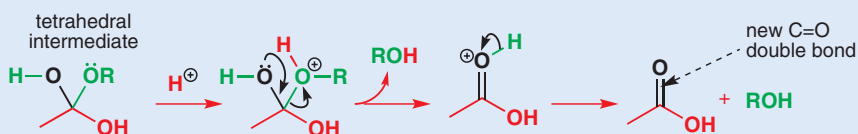
The HSO₄[−] anion is not involved in the rate-determining formation of the carbocation, and is also a very bad nucleophile, so it does not attack the C atom of the carbocation. Neither is it basic, but you can see from the mechanism that it does behave as a base (that is, it removes a proton). It does this only because it is even more feeble as a nucleophile. The rate equation will not involve the concentration of HSO₄[−], and the rate-determining step is the same as that in the S_N1 reaction—unimolecular loss of water from the protonated *t*-BuOH. This elimination mechanism is therefore called E1.

We will shortly come back to these two mechanisms for elimination, plus a third, but it is worth noting at this stage that the choice between E1 and E2 is not based on the same grounds as the choice between S_N1 and S_N2: you have just seen both E1 and E2 elimination from a substrate that would only undergo S_N1. The difference between the two reactions was the strength of the base, so first we need to answer the question: when does a nucleophile start behaving as a base?

Elimination in carbonyl chemistry

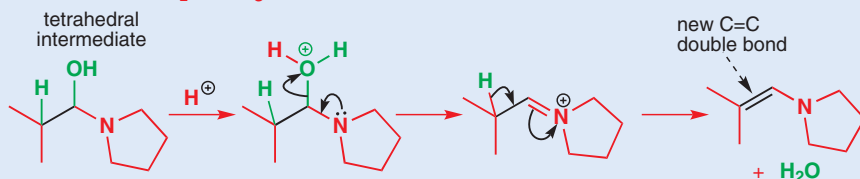
We have left detailed discussion of the formation of alkenes until this chapter, but we used the term ‘elimination’ in Chapters 10 and 11 to describe the loss of a leaving group from a tetrahedral intermediate. For example, the final steps of acid-catalysed ester hydrolysis involve E1 elimination of ROH to leave a double bond: C=O rather than C=C.

E1 elimination of ROH during ester hydrolysis



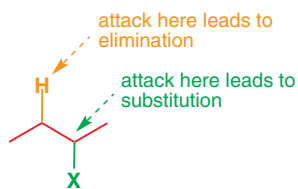
In Chapter 11 you even saw an E1 elimination giving an alkene. That alkene was an enamine—here is the reaction.

E1 elimination of H₂O during enamine formation



How the nucleophile affects elimination versus substitution

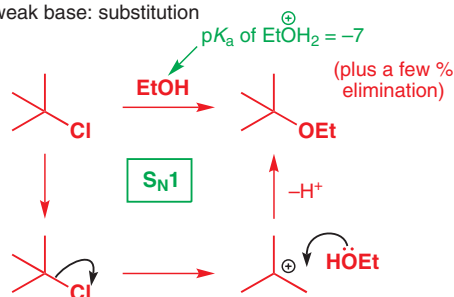
Basicity



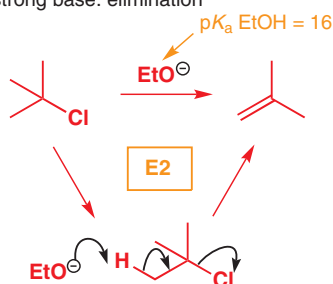
You have just seen molecules bearing leaving groups being attacked at two distinct electrophilic sites: the carbon to which the leaving group is attached, and the hydrogen atoms on the carbon adjacent to the leaving group. Attack at carbon leads to substitution; attack at hydrogen leads to elimination. Since strong bases attack protons, it is generally true that, the more basic the nucleophile, the more likely that elimination is going to replace substitution as the main reaction of an alkyl halide.

Here is an example of this idea at work: a weak base (EtOH) leads to substitution while a strong base (ethoxide ion) leads to elimination.

weak base: substitution



strong base: elimination



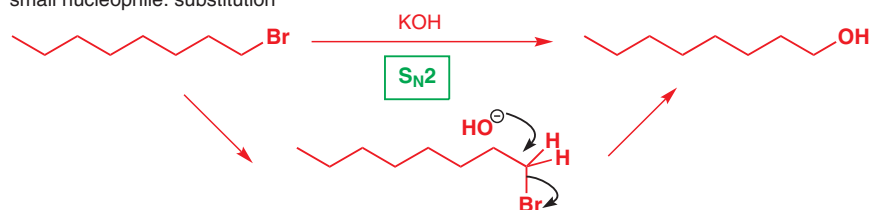
Elimination, substitution, and hardness

We can also rationalize selectivity for elimination versus substitution, or attack on H versus attack on C in terms of hard and soft electrophiles (p. 357). In an $\text{S}_{\text{N}}2$ substitution, the carbon centre is a soft electrophile—it is essentially uncharged, and with leaving groups such as halide the $\text{C}-\text{X} \sigma^*$ is a relatively low-energy LUMO. Substitution is therefore favoured by nucleophiles whose HOMOs are best able to interact with this LUMO—in other words soft nucleophiles. In contrast, the $\text{C}-\text{H} \sigma^*$ is higher in energy because the atoms are less electronegative. This, coupled with the hydrogen's small size, makes the $\text{C}-\text{H}$ bond a hard electrophilic site, and as a result hard nucleophiles favour elimination.

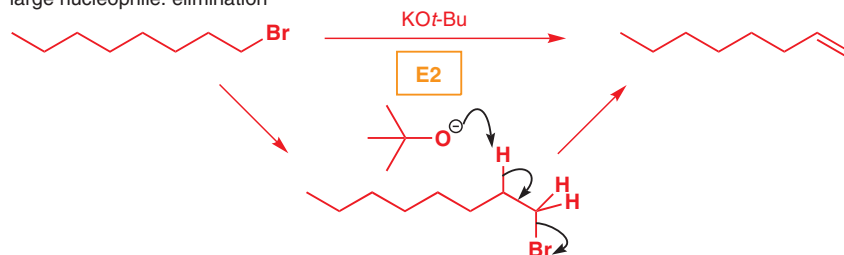
Size

For a nucleophile, attacking a carbon atom means squeezing past its substituents—and even for unhindered primary alkyl halides there is still one alkyl group attached. This is one of the reasons that $\text{S}_{\text{N}}2$ is so slow on hindered alkyl halides—the nucleophile has difficulty getting to the reactive centre. Getting at a more exposed hydrogen atom in an elimination reaction is much easier, and this means that, as soon as we start using basic nucleophiles that are also bulky, elimination becomes preferred over substitution, even for primary alkyl halides. One of the best bases for promoting elimination and avoiding substitution is potassium *t*-butoxide. The large alkyl substituent makes it hard for the negatively charged oxygen to attack carbon in a substitution reaction, but it has no problem attacking hydrogen.

small nucleophile: substitution



large nucleophile: elimination



Temperature

Temperature has an important role to play in deciding whether a reaction is an elimination or a substitution. In an elimination, two molecules become three (count them). In a substitution, two molecules form two new molecules. The two reactions therefore differ in the change in entropy during the reaction: ΔS is greater for elimination than for substitution. In Chapter 12, we discussed the equation

■ This explanation is simplified because what matters is the rate of the reaction, not the stability of the products. A detailed discussion is beyond the scope of the book, but the general argument still holds.

$$\Delta G = \Delta H - T\Delta S$$

This equation says that a reaction in which ΔS is positive becomes more favourable (ΔG becomes more negative) at higher temperature. Eliminations should therefore be favoured at high temperature, and this is indeed the case: most eliminations you will see are conducted at room temperature or above.

➡ For a related example see Chapter 12, p. 247

● Elimination versus substitution

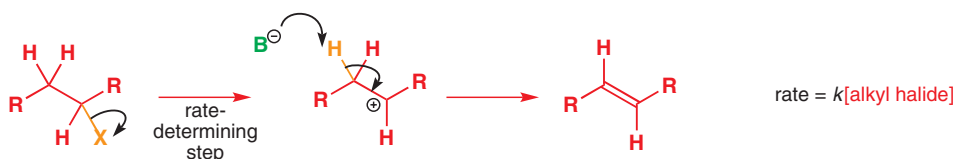
- Nucleophiles that are strong bases favour elimination over substitution.
- Nucleophiles (or bases) that are bulky favour elimination over substitution.
- High temperatures favour elimination over substitution.

E1 and E2 mechanisms

Now that you have seen a few examples of elimination reactions, it is time to return to our discussion of the two mechanisms for elimination. To summarize what we have said so far:

- E1 describes an elimination reaction (E) in which the rate-determining step is unimolecular (1) and does not involve the base. The leaving group leaves in this step, and the proton is removed in a separate second step.

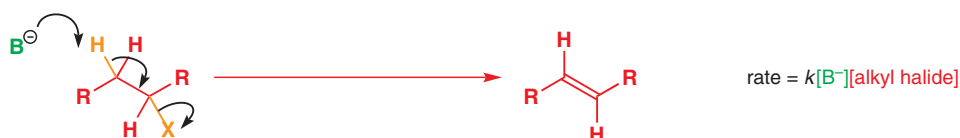
general mechanism for E1 elimination



■ In E2 eliminations the loss of the leaving group and removal of the proton are **concerted**.

- E2 describes an elimination (E) that has a bimolecular (2) rate-determining step that must involve the base. Loss of the leaving group is simultaneous with removal of the proton by the base.

general mechanism for E2 elimination



There are a number of factors that affect whether an elimination goes by an E1 or E2 mechanism. One is immediately obvious from the rate equations: only the E2 is affected by the concentration of base, so at high base concentration E2 is favoured. The rate of an E1 reaction is not even affected by what base is present—so E1 is just as likely with weak as with strong bases, while E2 goes faster with strong bases than weak ones: strong bases at whatever concentration will favour E2 over E1. If you see that a strong base is required for an elimination, it is certainly an E2 reaction. Take the first elimination in this chapter as an example.

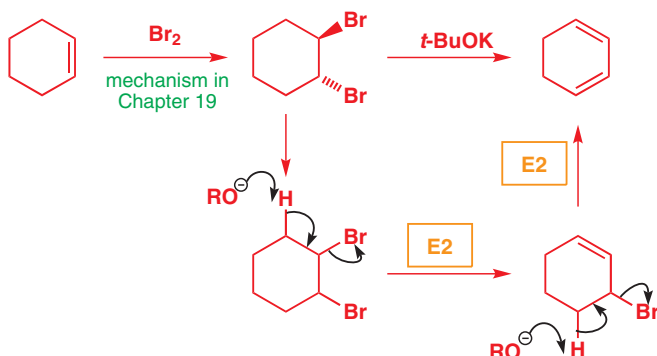
reaction of *t*-butyl bromide with concentrated hydroxide



With less hindered alkyl halides hydroxide would not be a good choice as a base for an elimination because it is rather small and still very good at S_N2 substitutions (and even with tertiary alkyl halides, substitution outpaces elimination at low concentrations of hydroxide). So what are good alternatives?

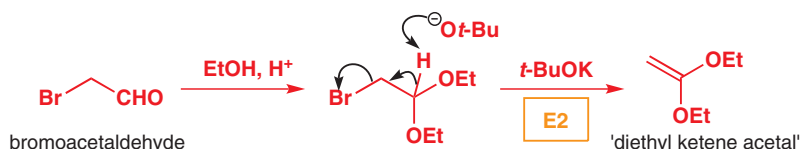
We have already mentioned the bulky *t*-butoxide—ideal for promoting E2 as it's both bulky and a strong base (pK_a of *t*-BuOH = 18). Here it is at work converting a dibromide to a diene with two successive E2 eliminations. Since dibromides can be made from alkenes (you will see how in the next chapter), this is a useful two-step conversion of an alkene to a diene.

synthesis of a diene by a double E2 elimination



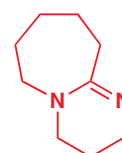
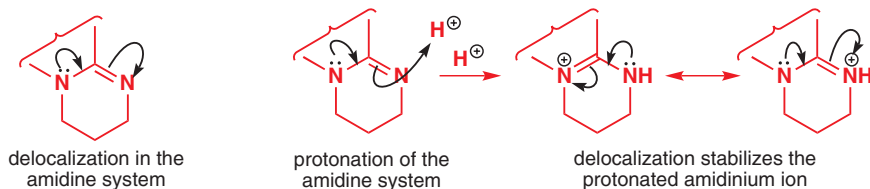
Interactive mechanism for double E2 to form diene

The product of the next reaction is a 'ketene acetal'. Unlike most acetals, this one can't be formed directly from ketene (ketene, $CH_2=C=O$, is too unstable), so instead the acetal is made by the usual method from bromoacetaldehyde and then HBr is eliminated using $t-BuOK$.



➡ You will meet ketene, briefly, in the next chapter.

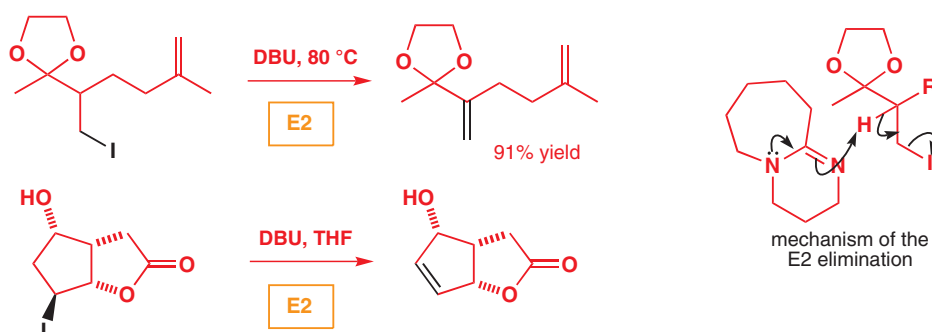
Among the most commonly used bases for converting alkyl halides to alkenes is one that you met in Chapter 8: DBU. This base is an amidine—delocalization of one nitrogen's lone pair onto the other, and the resulting stabilization of the protonated amidinium ion, makes it particularly basic, with a pK_a (of the protonated amidine) of about 12.5. There is not much chance of getting those voluminous fused rings into tight corners—so they pick off the easy-to-reach protons rather than attacking carbon atoms in substitution reactions.



DBU
1,8-diazabicyclo-
[5.4.0]undecene-7

➡ See p. 175 for more on DBU.

DBU will generally eliminate HX from alkyl halides to give alkenes. In these two examples, the products were used as intermediates in the synthesis of natural products.

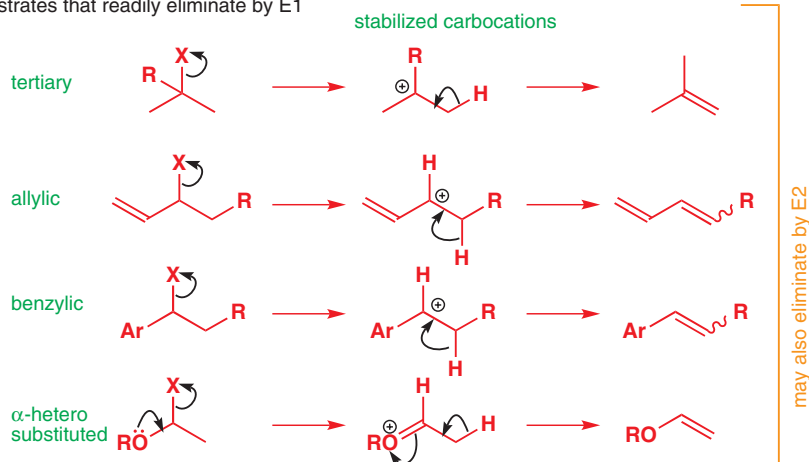


■ Note the use of high temperature to drive the elimination.

Substrate structure may allow E1

The first elimination of the chapter (*t*-BuBr plus hydroxide) illustrates something very important: the starting material is a tertiary alkyl halide and would therefore *substitute* only by S_N1 , but it can *eliminate* by either E2 (with strong bases) or E1 (with weak bases). The steric factors that disfavour S_N2 at hindered centres don't exist for eliminations. Nonetheless, E1 can occur *only* with substrates that can ionize to give relatively stable carbocations—tertiary, allylic, or benzylic alkyl halides, for example. Secondary alkyl halides may eliminate by E1, while primary alkyl halides only ever eliminate by E2 because the primary carbocation required for E1 would be too unstable. The chart below summarizes the types of substrate that can undergo E1—but remember that any of these substrates, under the appropriate conditions (in the presence of strong bases, for example), may also undergo E2. For completeness, we have also included in this chart three alkyl halides that cannot eliminate by either mechanism simply because they do not have any hydrogens to lose from carbon atoms adjacent to the leaving group.

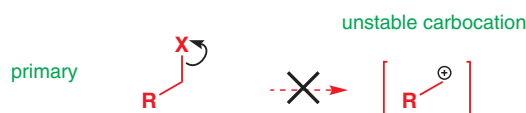
substrates that readily eliminate by E1



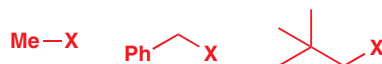
substrates that may eliminate by E1



substrates that never eliminate by E1



substrates that cannot eliminate by either mechanism—no appropriately placed hydrogens



cannot eliminate by E2

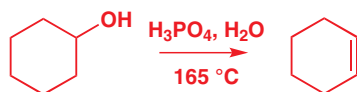
Can a proton just 'fall off' a cation?

In E1 mechanisms, once the leaving group has departed almost anything will serve as a base to remove a proton from the intermediate carbocation. Weakly basic solvent molecules (water or alcohols), for example, are quite sufficient, and you will often see the proton just 'falling off' in reaction mechanisms, with the assumption that there is a weak base somewhere to capture it. We showed the loss of a proton like this in the last example, and in the chart on this page.

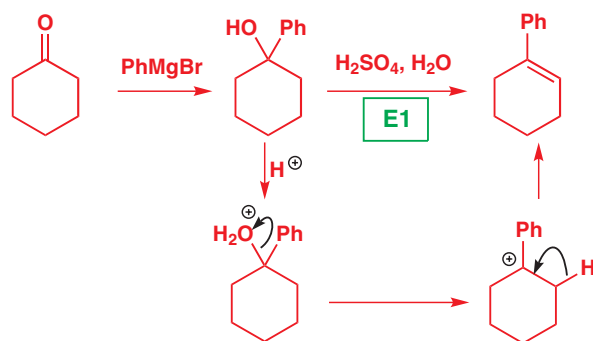


In very rare cases, such as the superacid solutions we described in Chapter 15 (p. 335), the cation is stable because counterions such as BF_4^- and SbF_6^- are not only non-nucleophilic but also so non-basic that they won't even accept a proton. This fact tells us that despite this common way of writing the E1 mechanism, *some* sort of weak base is necessary even for E1.

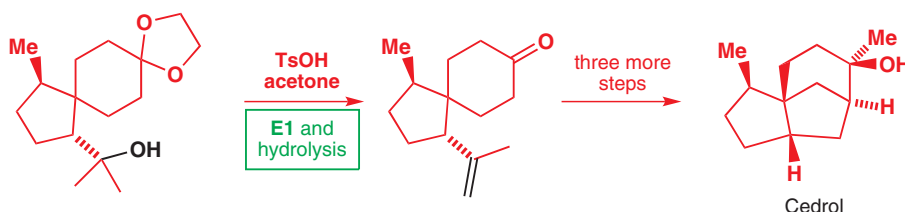
Polar solvents also favour E1 reactions because they stabilize the intermediate carbocation. E1 eliminations from alcohols in aqueous or alcohol solution are particularly common and very useful. An acid catalyst is used to promote loss of water, and in dilute H_2SO_4 , H_3PO_4 , or HCl the absence of good nucleophiles ensures that substitution does not compete. With phosphoric acid, for example, the secondary alcohol cyclohexanol gives cyclohexene.



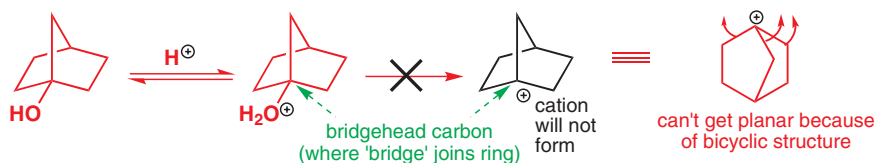
But the best E1 eliminations of all are with tertiary alcohols. The alcohols can be made using the methods of Chapter 9: nucleophilic attack by an organometallic on a carbonyl compound. Nucleophilic addition, followed by E1 elimination, is an excellent way of making this substituted cyclohexene, for example. Note that the proton required in the first step is recovered in the last—the reaction requires only catalytic amounts of acid.



Cedrol is important in the perfumery industry—it has a cedar wood fragrance. Corey's synthesis includes this step—the acid (toluenesulfonic acid, see p. 227) catalyses both the E1 elimination and the hydrolysis of the acetal.



At the end of the last chapter you met some bicyclic structures. These sometimes pose problems for elimination reactions. For example, this compound will not undergo elimination by either an E1 or an E2 mechanism.



We shall see shortly what the problem with E2 is, but for E1 the hurdle to be overcome is the formation of a planar carbocation. The bicyclic structure prevents the bridgehead carbon becoming planar so, although the cation would be tertiary, it is very high in energy and does not form. You could say that the non-planar structure forces the cation to have an empty sp^3 orbital instead of an empty p orbital, and we saw in Chapter 4 that it is always best to leave the orbitals with the highest possible energy empty.

➡ On p. 335 (Chapter 15) you saw a related example of an impossible $\text{S}_{\text{N}}1$ reaction with a non-planar cation.

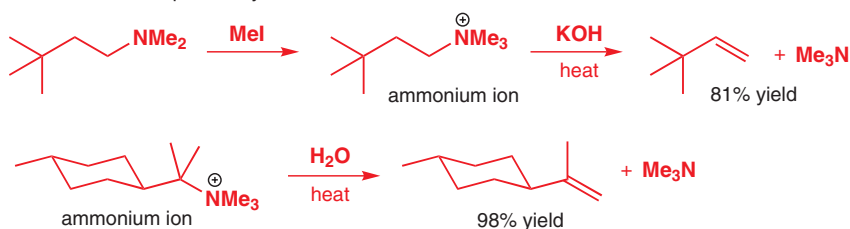
Bredt's rule

The impossibility of planar bridgehead carbons means that double bonds can almost never be formed to bridgehead carbons in bicyclic systems. This principle is known as Bredt's rule, but, as with all rules, it is much more important to know the reason than to know the name, and Bredt's rule is simply a consequence of the strain induced by a planar bridgehead carbon.

The role of the leaving group

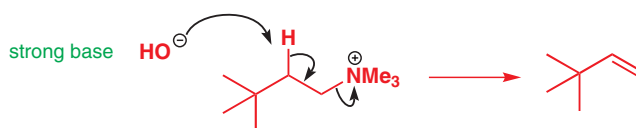
We haven't yet been very adventurous with our choice of leaving groups for eliminations: all you have seen so far are E2 from alkyl halides and E1 from protonated alcohols. This is deliberate: the vast majority of the two classes of eliminations use one of these two types of starting materials. But since the leaving group is involved in the rate-determining step of both E1 and E2, in general any good leaving group will lead to a fast elimination. You may, for example, see amines acting as leaving groups in eliminations of quaternary ammonium salts.

eliminations from quaternary ammonium ions

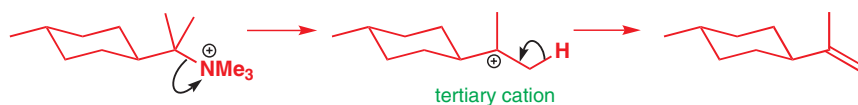


Both E1 and E2 are possible, and from what you have read so far you should be able to spot that there is one of each here: in the first example, a stabilized cation cannot be formed (so E1 is impossible), but a strong base is used, allowing E2. In the second, a stabilized tertiary cation could be formed (so *either* E1 or E2 might occur), but no strong base is present, so the mechanism must be E1.

E2 elimination



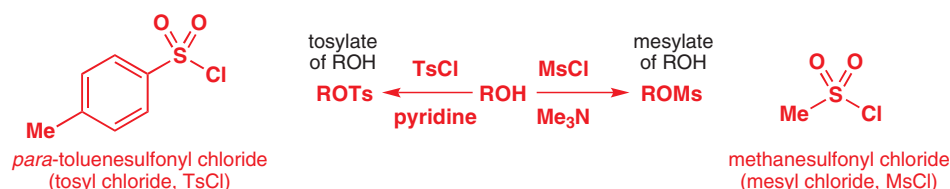
E1 elimination



You have just seen that hydroxyl groups can be turned into good leaving groups in acid, but this is only useful for substrates that can react by E1 elimination. The hydroxyl group is *never* a leaving group in E2 eliminations, since they have to be done in base. A strong base would remove the proton from the OH group instead.

● **OH⁻ is never a leaving group in an E2 reaction.**

For primary and secondary alcohols, the hydroxyl is best made into a leaving group for elimination reactions by sulfonation with *para*-toluenesulfonyl chloride (tosyl chloride, TsCl) or methanesulfonyl chloride (mesyl chloride, MeSO₂Cl or MsCl).



➡ You met the sulfonate esters—tosylates and mesylates—in Chapter 15 (p. 344).

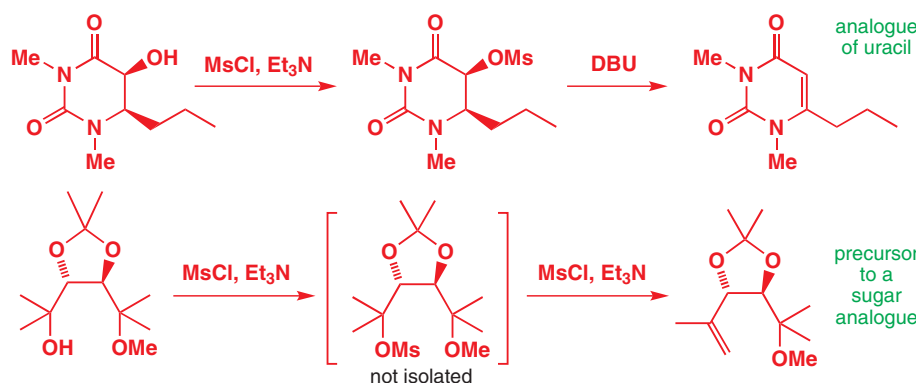
Toluenesulfonate esters (tosylates) can be made from alcohols (with TsCl, pyridine). We introduced tosylates in Chapter 15 because they are good electrophiles for substitution reactions with *non-basic* nucleophiles. With strong bases such as *t*-BuOK, NaOEt, or DBU they undergo very efficient elimination reactions. Here are two examples.

E2 eliminations of tosylates



Methanesulfonyl esters (or mesylates; Chapter 15) can be eliminated using DBU, but a good way of using MsCl to convert alcohols to alkenes is to do the mesylation and elimination steps in one go, using the same base (Et_3N) for both. Here are two examples making biologically important molecules. In the first, the mesylate is isolated and then eliminated with DBU to give a synthetic analogue of uracil, one of the nucleotide bases present in RNA. In the second, the mesylate is formed and eliminated in the same step using Et_3N , to give a precursor to a sugar analogue.

➡ There is more about RNA bases and sugars in Chapter 42.

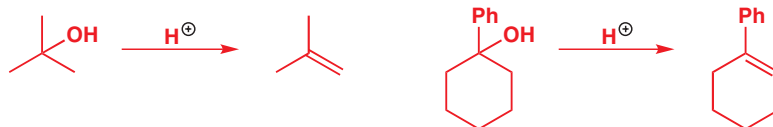


The second example here involves (overall) the elimination of a tertiary alcohol—so why couldn't an acid-catalysed E1 reaction have been used? The problem here, nicely solved by the use of the mesylate, is that the molecule contains an acid-sensitive acetal functional group. An acid-catalysed reaction would also have risked eliminating methanol from the other tertiary centre.

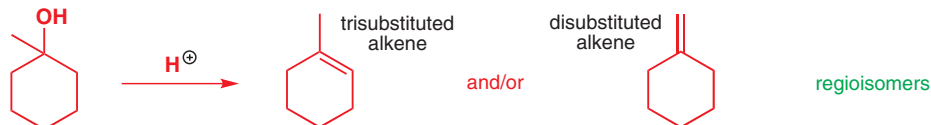
E1 reactions can be stereoselective

For some eliminations only one product is possible. For others, there may be a choice of two (or more) alkene products that differ either in the location or stereochemistry of the double bond. We shall now move on to discuss the factors that control the stereochemistry (geometry—*cis* or *trans*) and regiochemistry (that is, where the double bond is) of the alkenes, starting with E1 reactions.

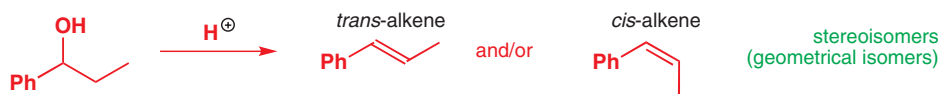
only one alkene possible



two regioisomeric alkenes possible

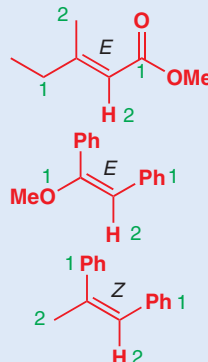


two stereoisomeric alkenes possible

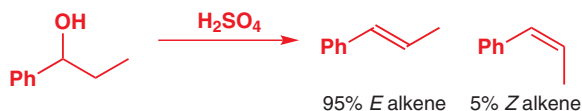


E and Z alkenes

You met the idea that alkenes can exist as geometrical *cis* and *trans* isomers in Chapters 3 and 7, and now that you have read Chapter 14 we can be more precise with our definitions. *cis* and *trans* are rather loosely defined terms (like *syn* and *anti*), although no less useful for it. But for formal assignment of geometry, we use the stereochemical descriptors *E* and *Z*. For disubstituted alkenes, *E* corresponds to *trans* and *Z* corresponds to *cis*. To assign *E* or *Z* to tri- or tetrasubstituted alkenes, the groups at either end of the alkene are given an order of priority according to the same rules as those outlined for *R* and *S* in Chapter 15. If the two higher priority groups are *cis*, the alkene is *Z*; if they are *trans* the alkene is *E*. Of course, molecules don't know these rules, and sometimes (as in the second example here) the *E* alkene is less stable than the *Z*.

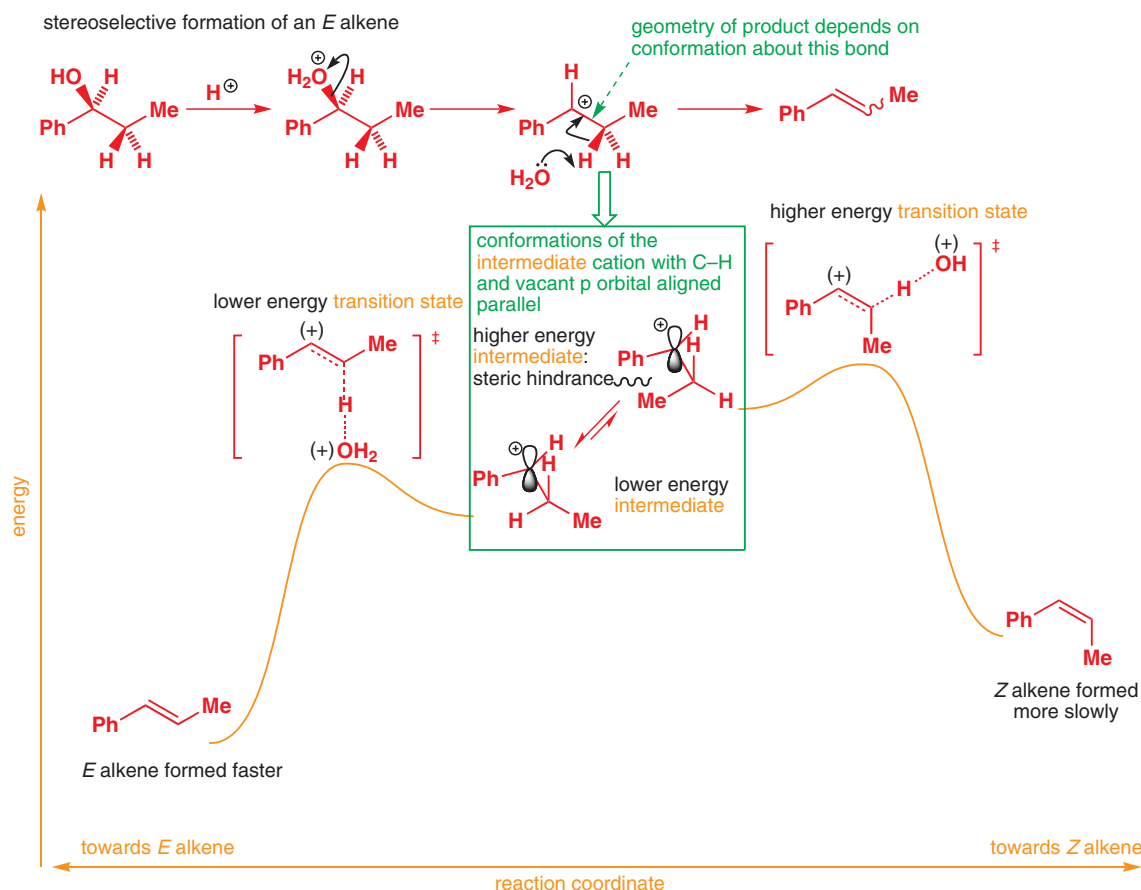


E alkenes (and transition states leading to *E* alkenes) are usually lower in energy than *Z* alkenes (and the transition states leading to them) for steric reasons: the substituents can get further apart from one another. A reaction that can choose which it forms is therefore likely to favour the formation of *E* alkenes. For alkenes formed by E1 elimination, this is exactly what happens: the less hindered *E* alkene is favoured. Here is an example.



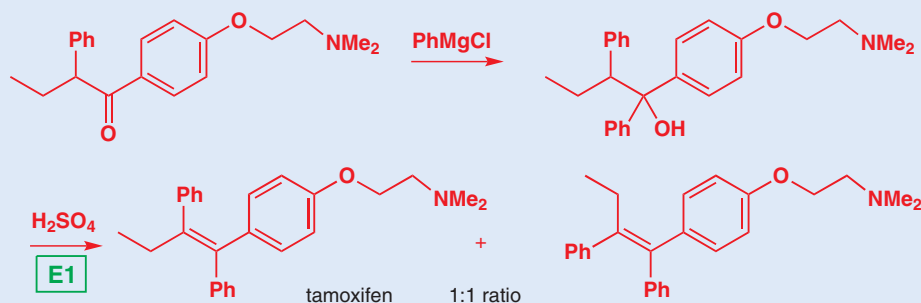
The geometry of the product is determined at the moment that the proton is lost from the intermediate carbocation. The new π bond can only form if the vacant p orbital of the carbocation and the breaking C–H bond are aligned parallel. In the example shown there are two possible conformations of the carbocation with parallel orientations, but one is more stable than the other because it suffers less steric hindrance. The same is true of the transition states on the route to the alkenes—the one leading to the *E* alkene is lower in energy, and more *E* alkene than *Z* alkene is formed. The process is stereoselective because the reaction chooses to form predominantly one of two possible stereoisomeric products.

➡ In Chapter 39 we shall discuss why the transition states for the decomposition of high-energy intermediates like carbocations are very similar in structure to the carbocations themselves.



Tamoxifen

Tamoxifen is an important drug in the fight against breast cancer, one of the most common forms of cancer. It works by blocking the action of the female sex hormone oestrogen. The tetrasubstituted double bond can be introduced by an E1 elimination: there is no ambiguity about where the double bond goes, although the two stereoisomers form in about equal amounts. Tamoxifen is the *Z* isomer.

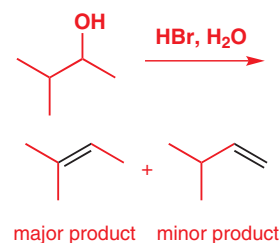


➡ We will come back to the most useful ways of controlling the geometry of double bonds in Chapter 27.

E1 reactions can be regioselective

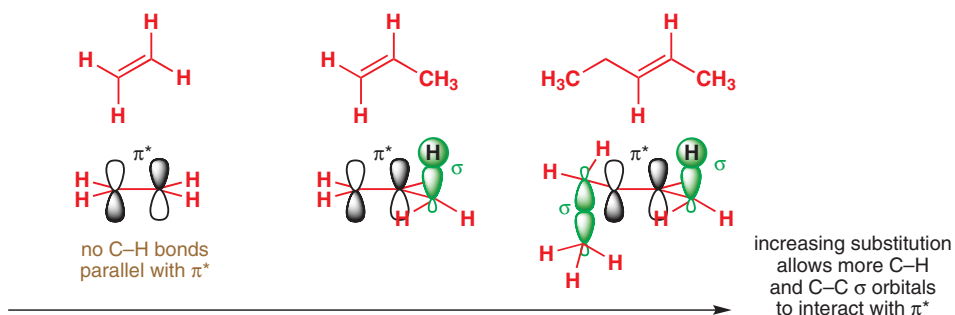
We can use the same ideas when we think about E1 eliminations that can give more than one regioisomeric alkene. Here is an example. The major product is the alkene that has the more substituents because this alkene is the more stable of the two possible products.

- More substituted alkenes are more stable.

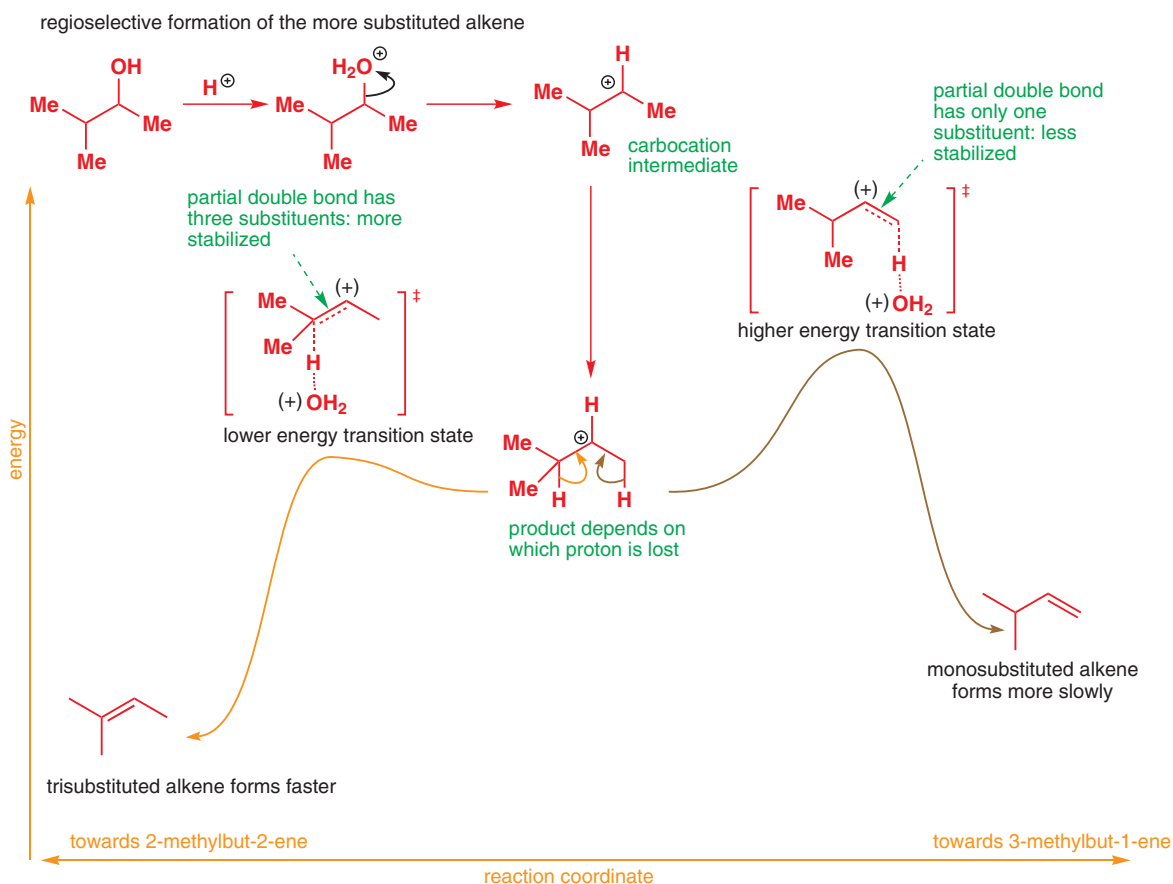


This is quite a general principle. But why should it be true? The reason for this is related to the reason why more substituted carbocations are more stable. In Chapter 15 we said that the carbocation is stabilized when its empty p orbital can interact with the filled orbitals of parallel C–H and C–C bonds. The same is true of the π system of the double bond—it is stabilized when the empty π^* antibonding orbital can interact with the filled orbitals of parallel C–H and C–C bonds. The more C–C or C–H bonds there are, the more stable the alkene.

■ This explanation of both stereo and regioselectivity in E1 reactions is based on *kinetic* arguments—which alkene forms faster. But it is also true that some E1 eliminations are reversible: the alkenes may be protonated in acid to re-form carbocations, as you will see in the next chapter. This reprotonation allows the more stable product to form preferentially under *thermodynamic* control. In any individual case, it may not be clear which is operating. However, with E2 reactions, which follow, only kinetic control applies: E2 reactions are never reversible.



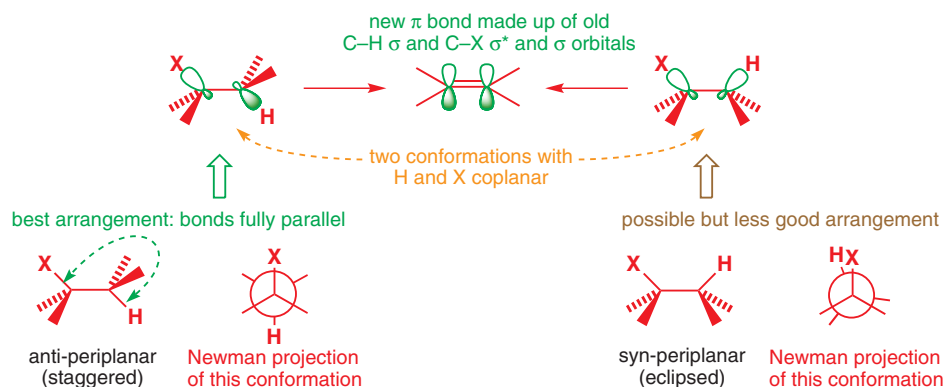
The more substituted alkene is more stable, but this does not necessarily explain why it is the one that forms faster. To do that, we should look at the transition states leading to the two alkenes. Both form from the same carbocation, but which one we get depends on which proton is lost. Removal of the proton on the right (brown arrow) leads to a transition state in which there is a monosubstituted double bond partly formed. Removal of the proton on the left (orange arrow) leads to a partial double bond that is trisubstituted. This is more stable—the transition state is lower in energy, and the more substituted alkene forms faster.



E2 eliminations have anti-periplanar transition states

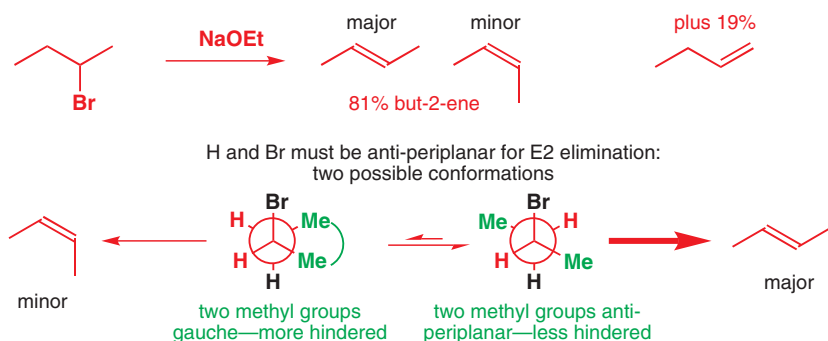
Although E1 reactions show some stereo and regioselectivity, the level of selectivity in E2 reactions can be much higher because of the more stringent demands on the transition state for E2 elimination. In an E2 elimination, the new π bond is formed by overlap of the C–H σ bond with the C–X σ^* antibonding orbital. The two orbitals have to lie in the same plane for best overlap, and now there are two conformations that allow this. One has H and X syn-periplanar, the other anti-periplanar. The anti-periplanar conformation is more stable because it is staggered (the syn-periplanar conformation is eclipsed) but, more importantly, only in the anti-periplanar conformation are the bonds (and therefore the orbitals) truly parallel.

➡ Look back to p. 365 if you need reminding of the shapes and names of the conformations of C–C single bonds.



➡ Newman projections illustrate the conformation of molecules viewed along the length of a bond. See p. 364 if you need reminding of how to draw and interpret them.

E2 eliminations therefore take place preferentially from the anti-periplanar conformation. We shall see shortly how we know this to be the case, but first we consider an E2 elimination that gives mainly one of two possible stereoisomers. 2-Bromobutane has two conformations with H and Br anti-periplanar, but the one that is less hindered leads to more of the product, and the *E* alkene predominates.



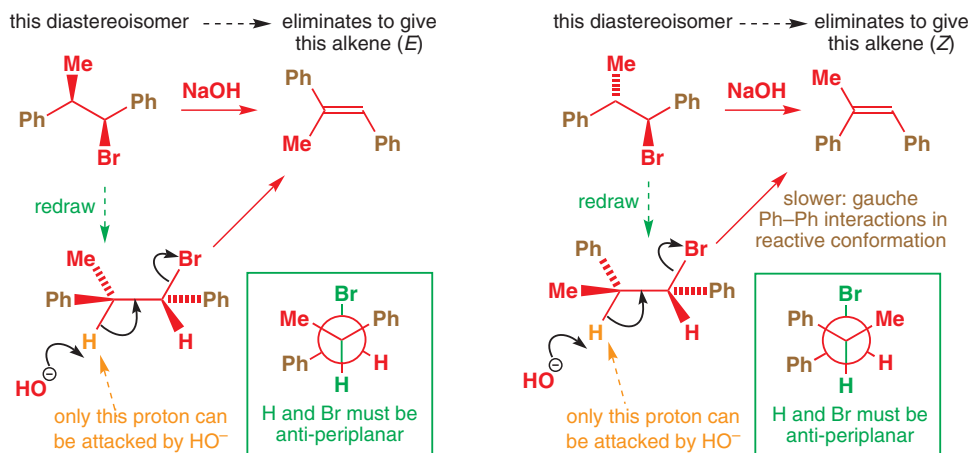
There is a choice of protons to be eliminated—the stereochemistry of the product results from which proton is anti-periplanar to the leaving group when the reaction takes place, and the reaction is stereoselective as a result.

Interactive mechanism for stereoselective E2

E2 eliminations can be stereospecific

In the next example there is only one proton that can take part in the elimination. Now there is no choice of anti-periplanar transition states. Whether the product is *E* or *Z*, the E2 reaction has only one course to follow. And the outcome depends on which diastereoisomer of the starting material is used. When the first diastereoisomer is drawn with the proton and

bromine anti-periplanar, as required, and in the plane of the page, the two phenyl groups have to lie one in front and one behind the plane of the paper. As the hydroxide attacks the C–H bond and eliminates Br^- , this arrangement is preserved and the two phenyl groups end up *trans* (the alkene is *E*). This is perhaps easier to see in the Newman projection of the same conformation.



Interactive mechanism for stereospecific E2

The second diastereomer forms the *Z* alkene for the same reasons: the two phenyl groups are now on the same side of the H–C–C–Br plane in the reactive anti-periplanar conformation (again, this is clear in the Newman projection) and so they end up *cis* in the product. Each diastereomer gives a different alkene geometry, and they do so at different rates. The first reaction is about ten times as fast as the second because, although this anti-periplanar conformation is the only reactive one, it is not necessarily the most stable. The Newman projection for the second reaction shows clearly that the two phenyl groups have to lie synclinal (*gauche*) to one another: the steric interaction between these large groups will mean that, at any time, a relatively small proportion of molecules will adopt the right conformation for elimination, slowing the process down.

Reactions in which the stereochemistry of the product is determined by the stereochemistry of the starting material are called **stereospecific**.

■ A stereospecific reaction is not simply a reaction that is very stereoselective! The two terms have different mechanistic meanings, and are not just different degrees of the same thing.

● Stereoselective or stereospecific?

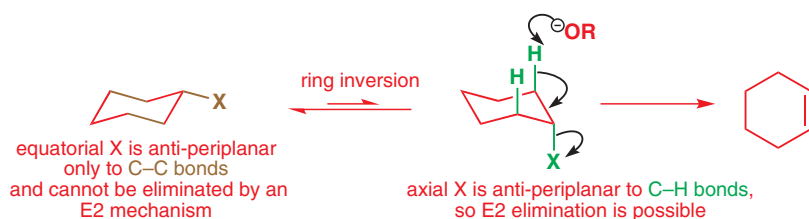
- **Stereoselective reactions** give one predominant product because the reaction pathway has a choice. Either the pathway of lower activation energy is preferred (kinetic control) or the more stable product is preferred (thermodynamic control).
- **Stereospecific reactions** lead to the production of a single isomer as a direct result of the mechanism of the reaction and the stereochemistry of the starting material. There is no choice. The reaction gives a different diastereoisomer of the product from each stereoisomer of the starting material.

E2 eliminations from cyclohexanes

The stereospecificity of the reactions you have just met is very good evidence that E2 reactions proceed through an anti-periplanar transition state. We know with which diastereoisomer we started, and we know which alkene we get, so there is no question over the course of the reaction.

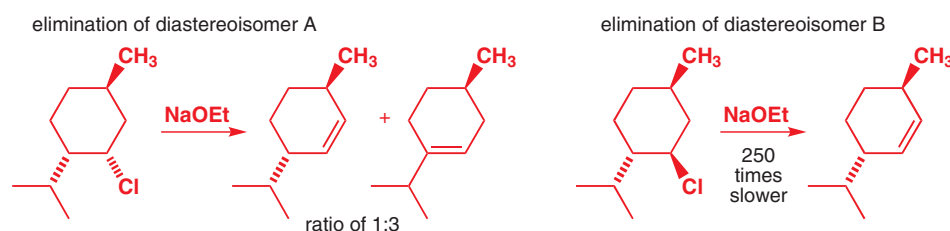
More evidence comes from the reactions of substituted cyclohexanes. You saw in Chapter 16 that substituents on cyclohexanes can be parallel with one another only if they are both axial. An equatorial C–X bond is anti-periplanar only to C–C bonds and cannot take part in an elimination. For mono-substituted cyclohexyl halides treated with base, this is not a problem because, although the axial conformer is less stable, there is still a significant amount present (see the table on p. 375), and elimination can take place from this conformer.

➡ In the next chapter (p. 415) you will see how the fact that pairs of axial bonds have overlapping orbitals also gives rise to distinctively large ^1H NMR coupling constants.



● For E2 elimination in cyclohexanes, both C-H and C-X must be axial.

These two diastereoisomeric cyclohexyl chlorides derived from menthol react very differently under the same conditions with sodium ethoxide as base. Both eliminate HCl but diastereoisomer A reacts rapidly to give a mixture of products, while diastereoisomer B (which differs only in the configuration of the carbon atom bearing chlorine) gives a single alkene product but very much more slowly. We can safely exclude E1 as a mechanism because the same cation would be formed from both diastereoisomers, and this would mean the ratio of products (although not necessarily the rate) would be the same for both.



The key to explaining reactions like this is to draw the conformation of the molecules. Both will adopt a chair conformation, and generally the chair having the largest substituent equatorial (or the largest *number* of substituents equatorial) is the more stable. In these examples the isopropyl group is most influential—it is branched and will have very severe 1,3-diaxial interactions if it occupies an axial position. In both diastereoisomers, an equatorial *i*-Pr also means an equatorial Me: the only difference is the orientation of the chlorine. For diastereoisomer A, the chlorine is forced axial in the major conformer: there is no choice because the relative configuration is fixed in the starting material. It's less stable than equatorial Cl, but is ideal for E2 elimination and there are two protons that are anti-periplanar available for removal by the base. The two alkenes are formed as a result of each of the possible protons with a 3:1 preference for the more substituted alkene.

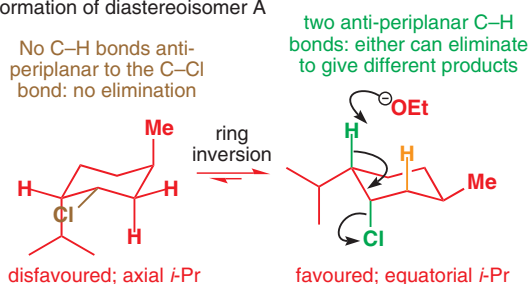
For diastereoisomer B, the chlorine is equatorial in the lowest-energy conformation. Once again there is no choice. But equatorial leaving groups cannot be eliminated by E2: in this conformation there is no anti-periplanar proton. This accounts for the difference in rate between the two diastereoisomers. A has the chlorine axial virtually all the time ready for E2, while B has an axial leaving group only in the minute proportion of the molecules that happen not to be in the lowest-energy conformation, but that have all three substituents axial. The all-axial conformer is much higher in energy, but only in this conformer can Cl[−] be eliminated. The concentration of reactive molecules is low, so the rate is also low. There is only one proton anti-periplanar and so elimination gives a single alkene.

■ This would be a good time to make sure you can reliably sketch a cyclohexane in the chair conformation. Our guidelines for helping you do this are on pp. 371–2.

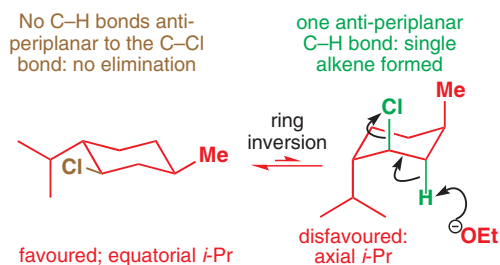
 Interactive mechanism for diastereoisomer A

 Interactive mechanism for diastereoisomer B

conformation of diastereoisomer A



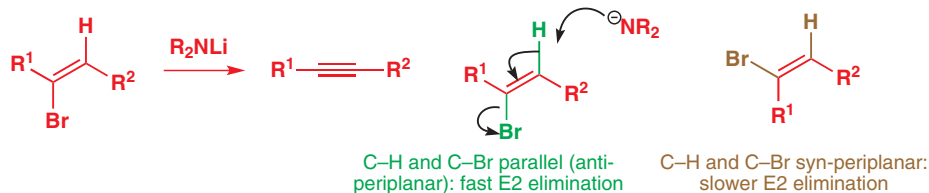
conformation of diastereoisomer B



E2 elimination from vinyl halides: how to make alkynes

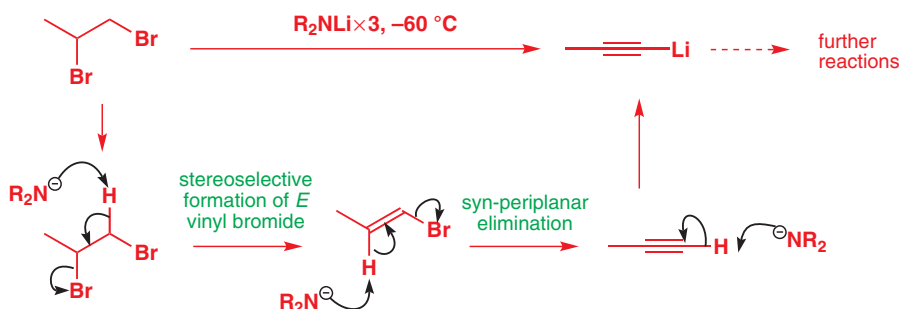
An anti-periplanar arrangement of C–Br and C–H is attainable with a vinylic bromide too, provided the Br and H are *trans* to one another. E2 elimination from the *Z* isomer of a vinyl bromide gives an alkyne rather faster than elimination from the *E* isomer because in the *E* isomer the C–H and C–Br bonds are syn-periplanar.

■ The base used here is LDA (lithium diisopropylamide) made by deprotonating *i*-Pr₂NH with BuLi (see p. 174). LDA is very basic (pK_a of *i*-Pr₂NH is about 35) but too hindered to be nucleophilic—ideal for promoting E2 elimination.



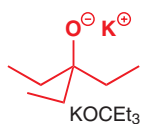
Vinyl bromides can themselves be made by elimination reactions of 1,2-dibromoalkanes. Watch what happens when 1,2-dibromopropane is treated with three equivalents of R₂NLi: first, elimination to the vinyl halide, then elimination of the vinyl halide to the alkyne. The terminal alkyne is amply acidic enough to be deprotonated by R₂NLi, and this is the role of the third equivalent. Overall, the reaction makes a lithiated alkyne (ready for further reactions) from a fully saturated starting material. This may well be the first reaction you have met that makes an alkyne from a starting material that doesn't already contain a triple bond.

making an alkyne from 1,2-dibromopropane



The regioselectivity of E2 eliminations

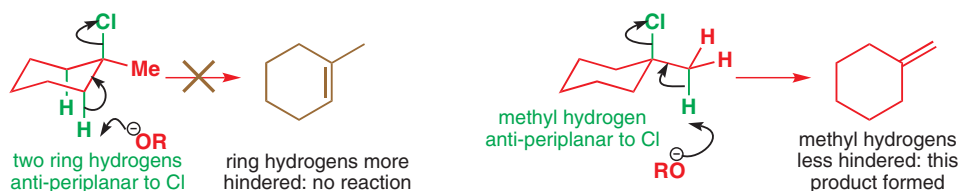
Here are two deceptively similar elimination reactions. The leaving group changes and the reaction conditions are very different but the overall process is elimination of HX to produce one of two alkenes.



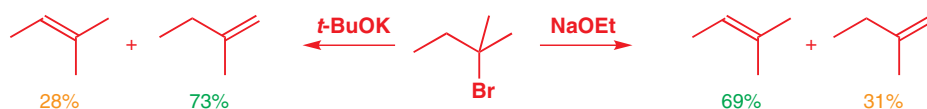
In the first example acid-catalysed elimination of water from a tertiary alcohol produces a trisubstituted alkene. Elimination of HCl from the corresponding tertiary alkyl chloride promoted by a very hindered alkoxide base (more hindered than *t*-BuOK because all the ethyl groups have to point away from one another) gives exclusively the less stable disubstituted alkene.

The reason for the two different regioselectivities is a change in mechanism. As we have already discussed, acid-catalysed elimination of water from tertiary alcohols is usually E1, and you already know the reason why the more substituted alkene forms faster in E1 reactions (p. 394). It should come as no surprise to you now that the second elimination, with a strong, hindered base, is an E2 reaction. But why does E2 give the less substituted product? This time, there is no problem getting C–H bonds anti-periplanar to the leaving group: in the conformation

with the Cl axial there are two equivalent ring hydrogens available for elimination, and removal of either of these would lead to the trisubstituted alkene. Additionally, any of the three equivalent methyl hydrogens are in a position to undergo E2 elimination to form the disubstituted alkene whether the Cl is axial or equatorial—and yet it is these and only these that are removed by the hindered base. The diagram summarizes two of the possibilities.



The base attacks the methyl hydrogens because they are less hindered—they are attached to a primary carbon atom, well away from the other axial hydrogens. E2 eliminations with hindered bases typically give the less substituted double bond because the fastest E2 reaction involves deprotonation at the least-substituted site. The hydrogens attached to a less substituted carbon atom are also more acidic. Think of the conjugate bases: a *t*-butyl anion is more basic (because the anion is destabilized by the three electron-donating alkyl groups) than a methyl anion, so the corresponding alkane must be less acidic. Steric factors are evident in the following E2 reactions, where changing the base from ethoxide to *t*-butoxide alters the major product from the more to the less substituted alkene.



● Elimination regioselectivity

- E1 reactions give the more substituted alkene.
- E2 reactions may give the more substituted alkene, but become more regioselective for the less substituted alkene with more hindered bases.

Hofmann and Saytsev

Traditionally, these two opposite preferences—for the more or the less substituted alkenes—have been called Saytsev's rule and Hofmann's rule, respectively. You will see these names used (along with a number of alternative spellings—acceptable for Saytsev, whose name is transliterated from Russian, but not for Hofmann: this Hofmann had one f and two n's), but there is little point remembering which is which (or how to spell them)—it is far more important to understand the reasons that favour formation of each of the two alkenes.

Anion-stabilizing groups allow another mechanism—E1cB

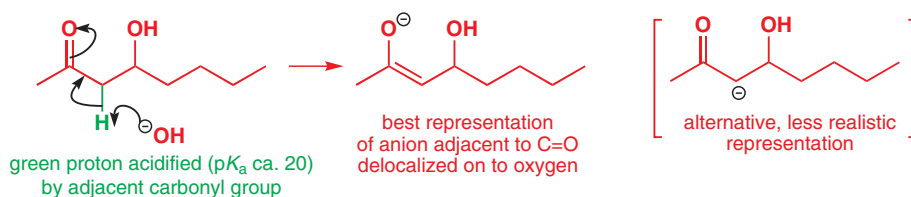
To finish this chapter, we consider a reaction that at first sight seems to go against what we have told you so far. It's an elimination catalysed by a strong base (KOH), so it looks like E2. But the leaving group is hydroxide, which we categorically (and truthfully) stated cannot be a leaving group in E2 eliminations.



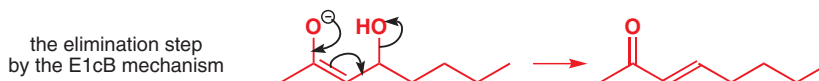
The key to what is going on is the carbonyl group. In Chapter 8 you met the idea that negative charges are stabilized by conjugation with carbonyl groups, and the list on p. 176 demonstrated how acidic a proton adjacent to a carbonyl group is. The proton that is removed in this elimination reaction is adjacent to the carbonyl group, and is therefore also rather acidic (pK_a about 20). This means that the base can remove it without the leaving group departing at the

■ This delocalized anion is called an **enolate**, and we will discuss enolates in more detail in Chapter 20 and beyond.

same time—the anion that results is stable enough to exist because it can be delocalized on to the carbonyl group.

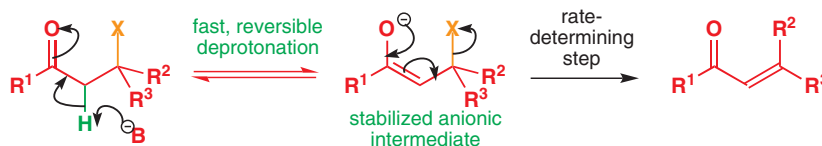


Although the anion is stabilized by the carbonyl group, it still prefers to lose a leaving group and become an alkene. This is the next step.



This step is also the rate-determining step of the elimination—the elimination is unimolecular and so is some kind of E1 reaction. The leaving group is not lost from the starting molecule, but from the *conjugate base* of the starting molecule, so this sort of elimination, which starts with a deprotonation, is called E1cB (cB for conjugate base). Here is the full mechanism, generalized for other carbonyl compounds.

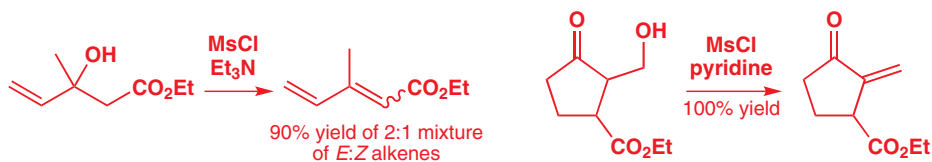
the E1cB mechanism



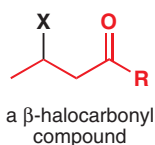
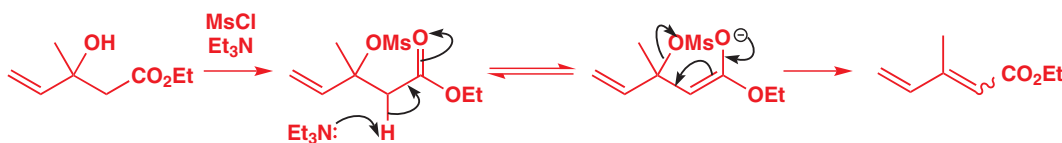
Interactive mechanism for E1cB elimination

■ 'E1cB' is written with no super- or subscripts, a lower-case c, and an upper-case B.

It's important to note that, while HO^- is never a leaving group in E2 reactions, it can be a leaving group in E1cB reactions. The anion it is lost from is already an alkoxide—the oxyanion does not need to be created as the HO^- is lost. The establishment of conjugation in the product also assists loss of HO^- . As the scheme above implies, other leaving groups are possible too. Here are two examples with methanesulfonate leaving groups.

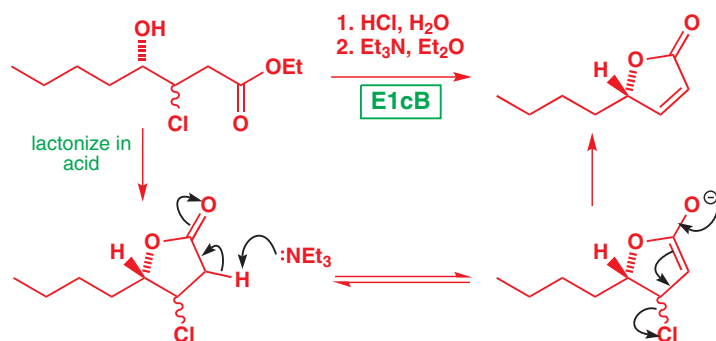


The first looks E1 (stabilized cation), the second E2—but in fact both are E1cB reactions. The most reliable way to spot a likely E1cB elimination is to see whether the alkene in the product is conjugated with a carbonyl group. If it is, the mechanism is probably E1cB.

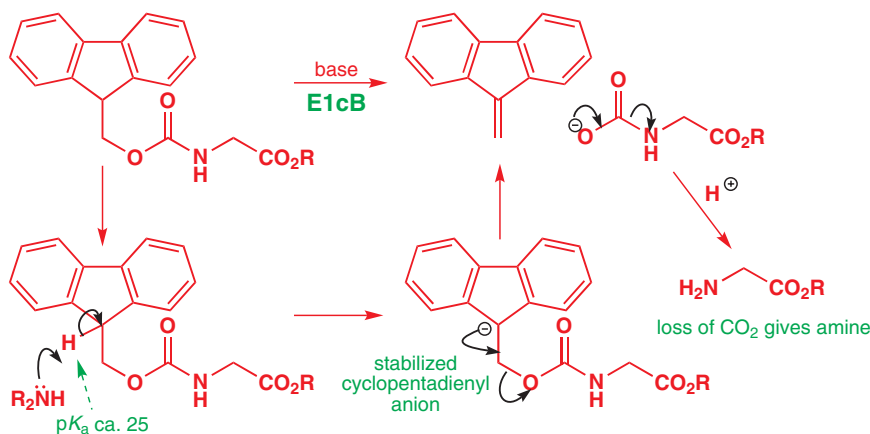


β-Halocarbonyl compounds can be rather unstable: the combination of a good leaving group and an acidic proton means that E1cB elimination is extremely easy. This mixture of diastereoisomers is first of all lactonized in acid (Chapter 10), and then undergoes E1cB

elimination with triethylamine to give a product known as a butenolide. Butenolides are common structures in naturally occurring compounds.



You will have noticed that we have shown the deprotonation step in the last few mechanisms as an equilibrium. Both equilibria lie rather over to the left-hand side because neither triethylamine (pK_a of Et₃NH⁺ about 10) nor hydroxide (pK_a of H₂O 15.7) is basic enough to remove completely a proton next to a carbonyl group ($pK_a > 20$). However, because the loss of the leaving group is essentially irreversible, only a small amount of deprotonated carbonyl compound is necessary to keep the reaction going. The important point about substrates that undergo E1cB is that there is some form of anion-stabilizing group next to the proton to be removed—it doesn't have to stabilize the anion very well but, as long as it makes the proton more acidic, an E1cB mechanism has a chance. Here is an important example with two phenyl rings helping to stabilize the anion, and a carbamate anion (R₂N—CO₂[−]) as the leaving group.

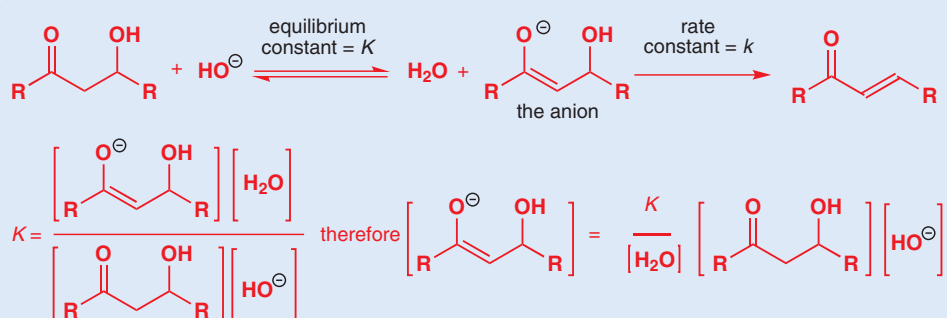


The proton to be removed has a pK_a of about 25 because its conjugate base is an aromatic cyclopentadienyl anion (we discussed this in Chapter 8). The E1cB elimination takes place with a secondary or tertiary amine as the base. Spontaneous loss of CO₂ from the eliminated product gives an amine, and you will meet this class of compounds again in Chapter 23, where we discuss the Fmoc protecting group.



The E1cB rate equation

The rate-determining elimination step in an E1cB reaction is unimolecular, so you might imagine it would have a first-order rate equation. In fact, the rate is also dependent on the concentration of base. This is because the unimolecular elimination involves a species—the anion—whose concentration is itself determined by the concentration of base by the equilibrium we have just been discussing. Using the following general E1cB reaction, the concentration of the anion can be expressed as shown.



The rate is proportional to the concentration of the anion, and we now have an expression for that concentration. We can simplify it further because the concentration of water is effectively constant.

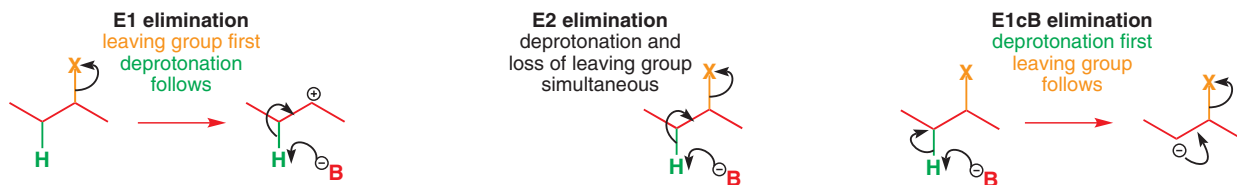
$$\text{rate} = k \frac{K}{[\text{H}_2\text{O}]} [\text{R}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}(\text{OH})-\text{R}][\text{HO}^-] = \text{constant} \times [\text{R}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}(\text{OH})-\text{R}][\text{HO}^-]$$

➡ You met this idea in Chapter 12 in the context of *third-order* rate equations.

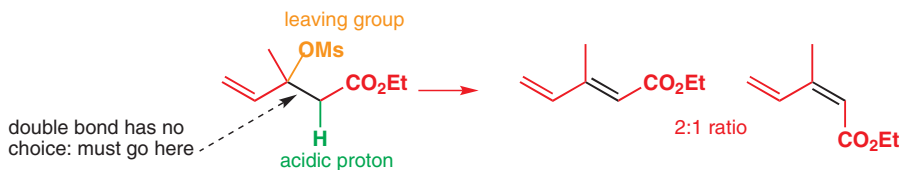
Just because the base (hydroxide) appears in this rate equation doesn't mean to say it is involved in the rate-determining step. Increasing the concentration of base makes the reaction go faster by increasing the amount of anion available to eliminate.

E1cB eliminations in context

We can also compare the E1cB mechanism with the other elimination reactions you have met by thinking of the relative timing of proton removal and leaving group departure. E1 is at one end of the scale: the leaving group goes first and proton removal follows in a second step. In E2 reactions, the two events happen at the same time: the proton is removed as the leaving group leaves. In E1cB the proton removal moves in front of leaving group departure.



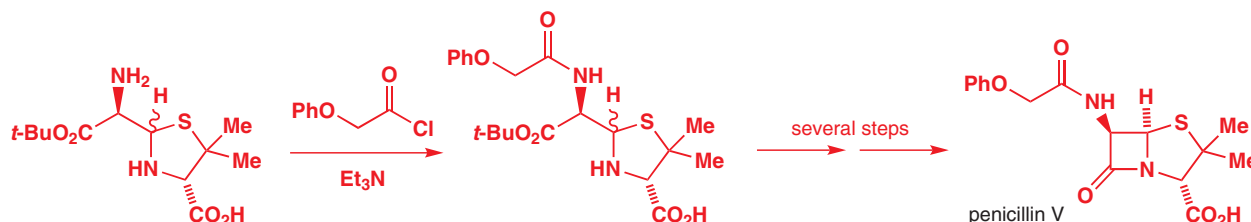
We talked about regio- and stereoselectivity in connection with E1 and E2 reactions. With E1cB, the regioselectivity is straightforward: the location of the double bond is defined by the position of (a) the acidic proton and (b) the leaving group.



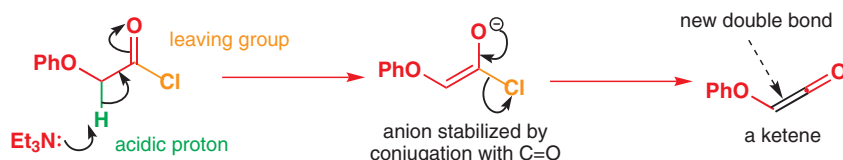
E1cB reactions may be stereoselective—the one above, for example, gives mainly the *E* alkene product (2:1 with *Z*). The intermediate anion is planar, so the stereochemistry of the starting materials is irrelevant, the less sterically hindered (usually *E*) product is preferred. This double E1cB elimination, for example, gives only the *E,E* product.



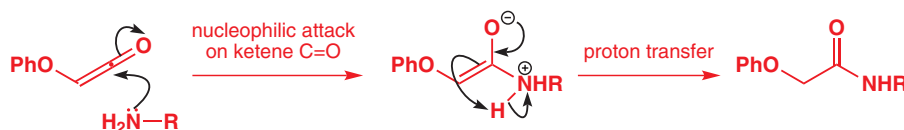
To finish this chapter we need to tell you about two E1cB eliminations that you may meet in unexpected places. We have saved them till now because they are unusual in that the leaving group is actually part of the anion-stabilizing group itself. First of all, try spotting the E1cB elimination in this step from the first total synthesis of penicillin V.



The reaction is deceptively simple—formation of an amide in the presence of base—and you would expect the mechanism to follow what we told you in Chapter 10. But the acyl chloride is, in fact, set up for an E1cB elimination—and you should expect this whenever you see an acyl chloride *with acidic protons next to the carbonyl group* used in the presence of a base such as triethylamine.



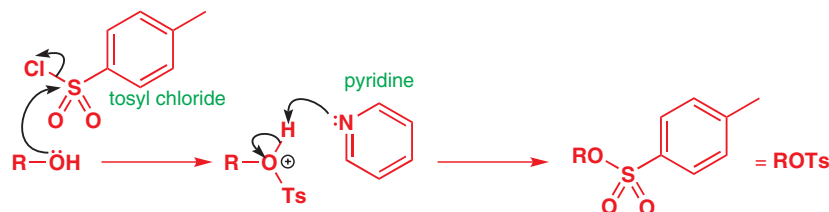
The product of the elimination is a substituted ketene—a highly reactive species whose parent structure is the molecule $\text{CH}_2=\text{C}=\text{O}$ that you will meet in the next chapter. It is the ketene that reacts with the amine to form the amide.



The second ‘concealed’ E1cB elimination is disguised in the mechanism of formation of methanesulfonates (mesylates). When we introduced sulfonate esters in Chapter 15, and revisited them on p. 391 of this chapter, we avoided (uncharacteristically, you may say) explaining the mechanism by which they are formed from sulfonyl chlorides. This was deliberate because, while TsCl reacts with alcohols by the mechanism you might predict, the reaction with MsCl involves an elimination step.

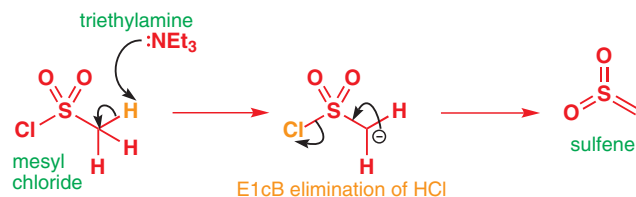
Here is the mechanism by which alkyl tosylates are formed from alcohols. The alcohol acts as a nucleophile towards the electrophilic sulfonyl chloride, and pyridine removes a proton to give the product.

formation of toluenesulfonates (tosylates): reagents $\text{ROH} + \text{TsCl} + \text{pyridine}$

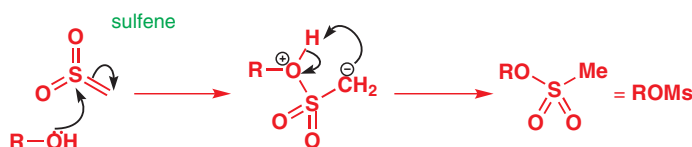


Methanesulfonyl chloride by contrast has a feature it shares with the acyl chlorides just above: a relatively acidic proton that can be removed by base. This deprotonation, followed by loss of chloride, is the first step in the formation of a mesylate ester. It is an E1cB elimination and the product is called a sulfene.

formation of methanesulfonates (mesylates): reagents ROH + MsCl + triethylamine



The sulfene is electrophilic in a slightly odd way: the alcohol acts as a nucleophile for sulfur and generates an anion of carbon which undergoes a proton transfer to give the mesylate. It is not uncommon for anions to form adjacent to sulfur, as you will see again in Chapter 27. Notice how similar the overall mechanism is to the acylation mechanism we showed you above.



To conclude

We finish with brief summaries of three important discussions we have had in this chapter.

Elimination *versus* substitution

The table below summarizes the general pattern of reactivity expected from various structural classes of alkyl halides (or tosylates, mesylates) in reactions with a representative range of nucleophiles (which may behave as bases).

		Poor nucleophile (e.g. H ₂ O, ROH)	Weakly basic nucleophile (e.g. I ⁻ , RS ⁻)	Strongly basic, unhindered nucleophile (e.g. RO ⁻)	Strongly basic, hindered nucleophile (e.g. DBU, <i>t</i> -BuO ⁻)
methyl	H₃C-X	no reaction	S _N 2	S _N 2	S _N 2
primary (unhindered)		no reaction	S _N 2	S _N 2	E2
primary (hindered)		no reaction	S _N 2	E2	E2
secondary		S _N 1, E1 (slow)	S _N 2	E2	E2
tertiary		E1 or S _N 1	S _N 1, E1	E2	E2
β to anion-stabilizing group		E1cB	E1cB	E1cB	E1cB

Some points about the table:

- Methyl halides cannot eliminate as there are no appropriately placed protons.
- Increasing branching favours elimination over substitution and strongly basic hindered nucleophiles always eliminate unless there is no option.
- Good nucleophiles undergo substitution by S_N2 unless the substrate is tertiary and then the intermediate cation can eliminate by E1 as well as substitute by S_N1 .
- High temperatures favour elimination by gearing up the importance of entropy in the free energy of reaction ($\Delta G = \Delta H - T\Delta S$). This is a good way of ensuring E1 in ambiguous cases.

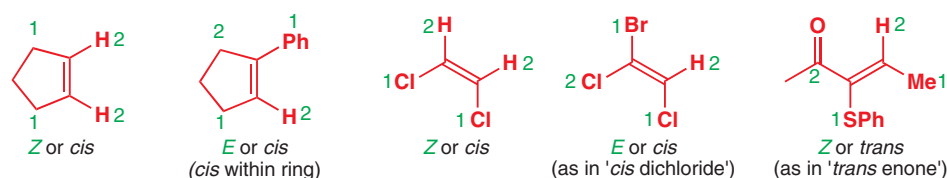
Summary of the stabilities of types of alkene

Alkenes are stabilized by:

- **conjugation**—anything that can conjugate with an alkene stabilizes it, including carbonyl groups, nitriles, benzene rings, RO or RNH groups, or another alkene. This is the strongest stabilization and usually dominates.
- **substitution**—alkyl groups stabilize alkenes weakly by σ -conjugation, so the more alkyl groups the better—but beware of the next point.
- **lack of steric hindrance**—as alkenes are planar, large, especially branched, substituents arranged *syn* on the alkene destabilize it, so tetra-substituted alkenes are usually less stable than tri-substituted ones. If the alkene is in a stable ring, this does not apply as the ring substituents have to be *syn* for the ring to exist.

Alkene stereochemistry: a summary of terminology

The official way of assigning alkene geometry is *E* and *Z*. *Z* comes from the German *zusammen* (together) and means that the two highest ranking substituents (by the same rules introduced in Chapter 14 for *R* and *S*) are on the same side of the alkene. The letter *Z* is a particularly unfortunate choice as it looks like a *trans* alkene! *E* comes from the German *entgegen* (opposed) and means that the two highest ranking substituents are on opposite sides (and, if anything also unfortunately looks like a *cis* alkene). The green numbers in the structures below show the relative rankings of the two substituents at each end of the alkene and the consequent assignment of geometry.



But possibly the most common method of referring to alkene geometry is to use *cis* and *trans*. These require a diagram as they refer to two substituents being on the same (*cis*) or opposite (*trans*) sides of the alkene. There is no specified order of priority and the speaker chooses substituents that are significant to the structure or to the reaction under discussion, making this a more flexible and versatile way of talking about alkenes. We have assigned *cis* and *trans* to the alkenes above to indicate their most important features, but notice the ambiguities that may occur.

■ This terminology may be used only for alkenes and not for three-dimensional stereochemistry.

■ Much the same are the terms *syn* and *anti*, introduced on p. 317 and used for relative three-dimensional stereochemistry. There is no formal definition, and a diagram is needed for clarity.

Further reading

See J. Keeler and P. Wothers, *Why Chemical Reactions Happen*, OUP, Oxford, 2003, chapter 11 for a comparison between substitution and elimination and F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry A, Structure and Mechanisms*, 5th edn, Springer 2007, chapter 5.

DBU and other strong bases are described in T. Ishikawa, ed. *Superbases for organic synthesis: guanidines, amidines and phosphazenes and related organocatalysts*, Wiley, Chichester, 2009. The trityl protecting group, along with many others is described in P. J. Kocienski, *Protecting Groups*, 3rd edn, Thieme, 2003.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>